Lab-2: Problem Identification and Feasibility Analysis

1. Title of the Project

Municipal Complaint Management System (MCMS)

2. Problem Statement & Identification

2.1 Background

Municipal corporations are responsible for delivering essential civic amenities such as sanitation, road maintenance, streetlight repairs, and water supply. Currently, citizens rely on manual methods—such as visiting municipal ward offices in person, writing physical complaint letters, or calling non-responsive helplines—to register grievances.

2.2 Key Issues in the Existing System

- **Lack of Transparency & Tracking:** Once a complaint is submitted physically, citizens have no visibility into its status, the officer assigned, or the estimated resolution time.
- **Manual Bottlenecks & Data Loss:** Paper-based complaint logs are prone to physical misplacement, data corruption, and deliberate delays by intermediaries.
- **Absence of Accountability & SLAs:** There is no strict Service Level Agreement (SLA) tracking or escalation mechanism for field workers and contractors who fail to resolve complaints on time.
- **Lack of Centralized Data & Analytics:** Municipal administrators and city commissioners lack actionable dashboards to identify chronic civic problem zones (e.g., recurring potholes or garbage dumps in specific wards).
- **Communication Gap:** Citizens are rarely notified when a grievance is resolved or closed, leading to repeated follow-ups and public dissatisfaction.

2.3 Proposed Solution

A web-based **Municipal Complaint Management System** that provides:

1. Online citizen registration and geotagged grievance lodging with image proof.
2. Automated departmental routing (Sanitation, Road Work, Electricity, Water Board).
3. Role-based dashboards for Citizens, Municipal Officers, and Field Workers.

4. End-to-end status tracking with a unique Tracking ID and citizen feedback upon resolution.

3. Feasibility Analysis

Feasibility study evaluates whether the proposed system is viable, practical, and cost-effective across five major dimensions (TELOS framework).

3.1 Technical Feasibility

- **Technology Stack:** The application utilizes mature, standard open-source web technologies:
    - *Frontend:* HTML5, CSS3/Tailwind CSS, JavaScript (React.js).
    - *Backend:* Node.js (Express) or Python (Django/FastAPI).
    - *Database:* Relational DBMS like PostgreSQL or MySQL.
- **Hardware Availability:** Does not require proprietary hardware. Can run on standard cloud infrastructure (AWS/GCP/Heroku) or local on-premise government servers.
- **Technical Expertise:** Standard web and database development skills are well within the capability of the software engineering team.
- **Conclusion:** The project is **Technically Feasible**.

3.2 Operational Feasibility

- **User Accessibility:** The citizen portal features a clean, responsive, and intuitive graphical user interface (GUI), requiring minimal technical or computer literacy from the general public.
- **Organizational Fit:** The system mirrors existing municipal administrative hierarchies (Officers -> Supervisors -> Field Workers), ensuring smooth operational adoption without disrupting established governance workflows.
- **Community Acceptance:** Citizens gain the convenience of lodging grievances 24/7 from home rather than visiting government offices during working hours.
- **Conclusion:** The project is **Operationally Feasible**.

3.3 Economic Feasibility

- **Development Cost:** The project relies on free, open-source software (FOSS), frameworks, and database engines, keeping direct software development costs minimal.
- **Operational Savings:**
    - Eliminates physical stationery, paper logs, and manual register filing.
    - Reduces administrative overhead and manual record management staffing costs.
- **Cost-Benefit Ratio:** The initial server deployment and maintenance costs are negligible compared to the significant long-term productivity gains and rapid grievance resolution.
- **Conclusion:** The project is **Economically Feasible**.

3.4 Schedule (Time) Feasibility

- **Project Breakdown:** The development follows an iterative software engineering lifecycle divided into standard lab milestones:
    - *Milestone 1:* Requirements Gathering & SRS.
    - *Milestone 2:* UML Analysis & System Design.
    - *Milestone 3:* Database Schema & API Prototyping.
    - *Milestone 4:* Integration, Testing, and Final Review.
- **Delivery Horizon:** The modular distribution among team members ensures the project can be completed within the allocated academic semester timeline.
- **Conclusion:** The project is **Schedule Feasible**.

3.5 Legal and Regulatory Feasibility

- **Data Privacy:** Stores minimal citizen data (Name, Phone/Email, Location) adhering to standard digital privacy and IT policies.
- **Compliance:** Complies with public electronic governance standards and does not violate any municipal copyright or regulatory frameworks.
- **Conclusion:** The project is **Legally Feasible**.

4. Final Feasibility Conclusion

The Feasibility Study confirms that the **Municipal Complaint Management System** is technically viable, economically cost-effective, operationally beneficial for citizens and municipal staff, and realistically executable within the project timeframe. Hence, proceeding to the Requirements Engineering and Design phases is fully justified.